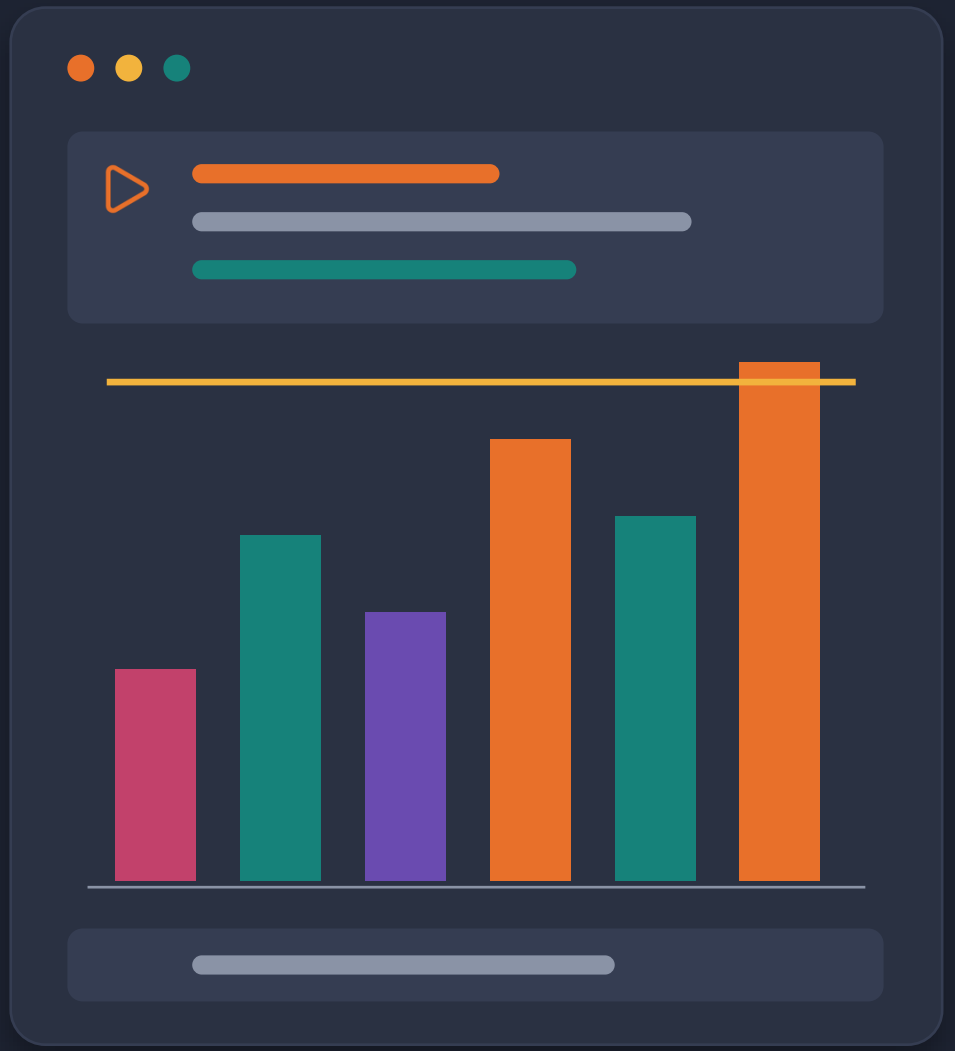
Lubin AI Workshop: Data Analysis

With Dr. Vishal Lala



Outline



How LLMs analyze data, reusable code, and why Python



Exercise 1: Baby Names



Daily sales by category: run the same script on each day's file



Exercise 2: AI job salaries



Interactive platforms, Colab setup, and the analysis workflow



Exercise 3: Bakery Transactions

How LLMs analyze data



You upload a spreadsheet and ask: “Analyze this data.”



The LLM uses learned knowledge to interpret the request and propose a method.



With execution tools, it can generate code and run calculations on the file.



It uses the returned outputs to explain the results.



Which data and calculations produced the answer?

Ask for the formula (i.e.,code) to do the analysis. But, Why?

Visibility:

A report may omit the method, filters, or exclusions.

Repeatability:

Regenerating an analysis may change the procedure.

Time:

Saved code can avoid repeated upload and model-response steps.

Model usage:

Reruns without AI calls need no additional LLM tokens.

Large files:

Uploads can encounter limits or incomplete coverage.

Automation:

Code can process new files without repeating a conversation.

LLMS · REUSABLE CODE · PYTHON

Prompt Once, Compute Many Times

1

Ask the LLM to generate code for a clearly defined analytical job.

2

Inspect the code, execute it, and validate the results.

3

Save the procedure and rerun it on compatible inputs.

Develop and refine the method with AI. Reuse the validated code when the job recurs.

Today's artifact: a notebook with visible code, checked outputs, and editable inputs.

Why Python?

Widely used:
A large community provides documentation, examples, and help.

Readable syntax:
Relatively concise code makes analytical steps easier to inspect.

AI support:
LLMs can generate, explain, and help debug Python code.

Data analysis libraries:
pandas handles tables, and Matplotlib creates charts.

Notebook workflow:
Run one cell at a time, inspect the output, and reuse the code.

We will use Python to turn a recurring daily sales question into a saved procedure.

DAILY SALES BY CATEGORY

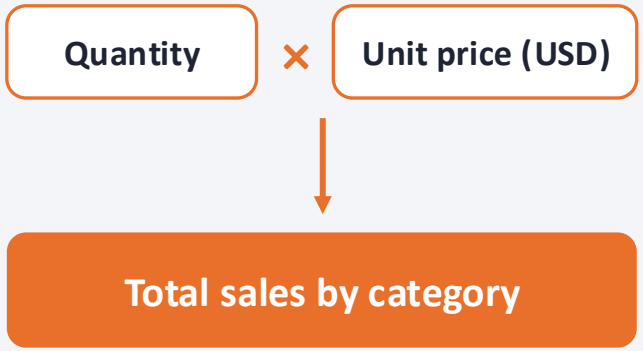
The problem, daily sales by category

Everyday Desk receives a new orders CSV each day.

Task: Find total sales in dollars for each product category, every day.

First five rows of orders_2026-10-26.csv, showing the columns used in the calculation:

Product	Category	Quantity	Unit price (USD)
USB-C Cable	Tech	1	$9.95
Water Bottle	Drinkware	1	$18.95
Wireless Mouse	Tech	1	$22.95
Water Bottle	Drinkware	1	$18.95
Notebook	Stationery	1	$7.50



DAILY SALES BY CATEGORY

The solution, run the saved Python script

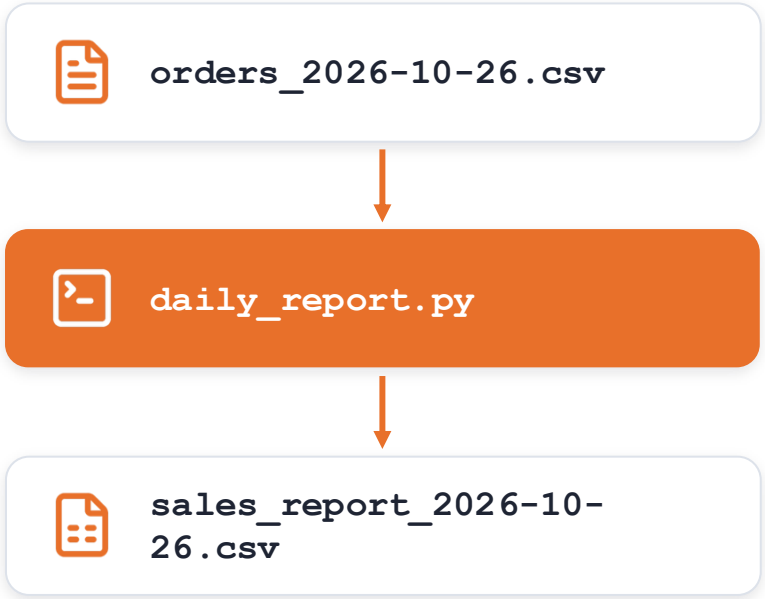
Run daily_report.py on the daily orders file:

```
python daily_report.py "orders_2026-10-26.csv"
```

The script calculates total sales by category and saves `sales_report_2026-10-26.csv`.

For the next day, change only the input filename:

```
python daily_report.py "orders_2026-10-27.csv"
```



The same code produces each day's category totals without another LLM request.

PLATFORMS · COLAB · WORKFLOW

Platforms for interactive AI-assisted analysis

The saved procedure can run in several environments. We will practice in Google Colab.

Platform	Working environment
Cursor	AI code editor with Python and notebook support
Google Antigravity	Agent-based coding environment with execution tools. FIFA Paradox analysis was done with Antigravity
VS Code	Python and Jupyter notebooks with AI assistance such as Copilot
JupyterLab + Jupyter AI	Jupyter notebooks with configurable AI chat and notebook tools
Google Colab	**Python notebooks in a browser with hosted execution**

Why Google Colab for this session?

Minimal setup:
No local Python installation for the hosted runtime.

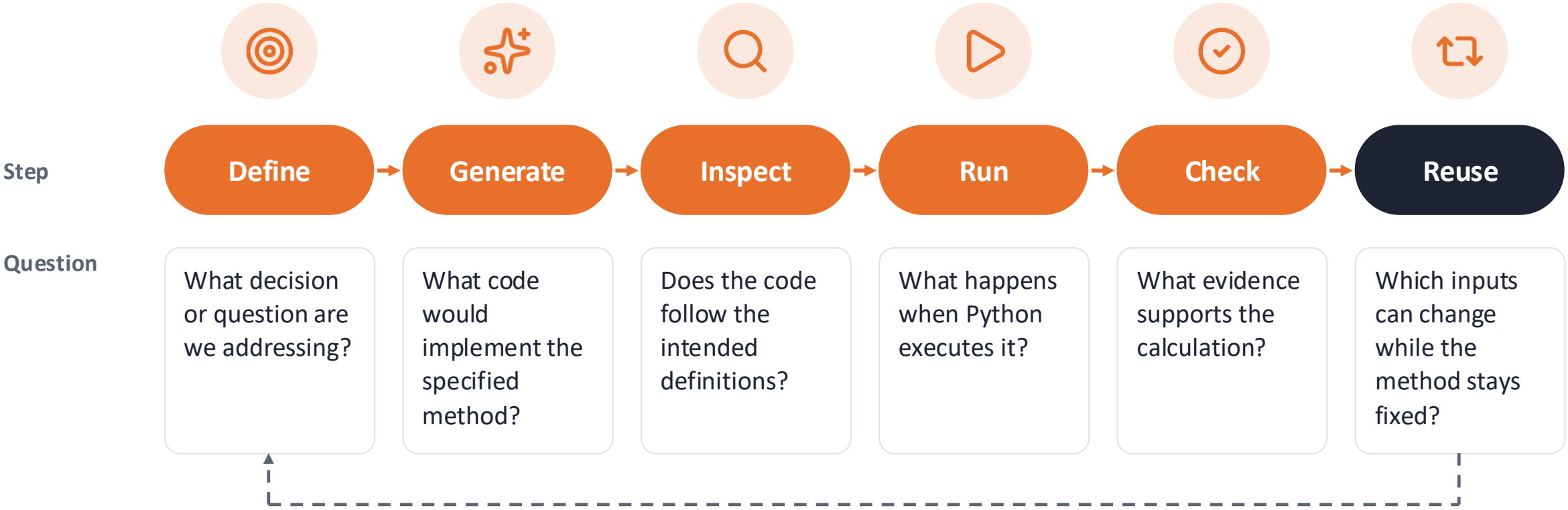
Visible steps:
Generate, inspect, and run code one cell at a time.

Accessible computing:
A free compute tier for introductory exercises.

Reusable notebooks:
Save, share, and download code with its outputs.

The analytical workflow

Use the same cycle in all three exercises.



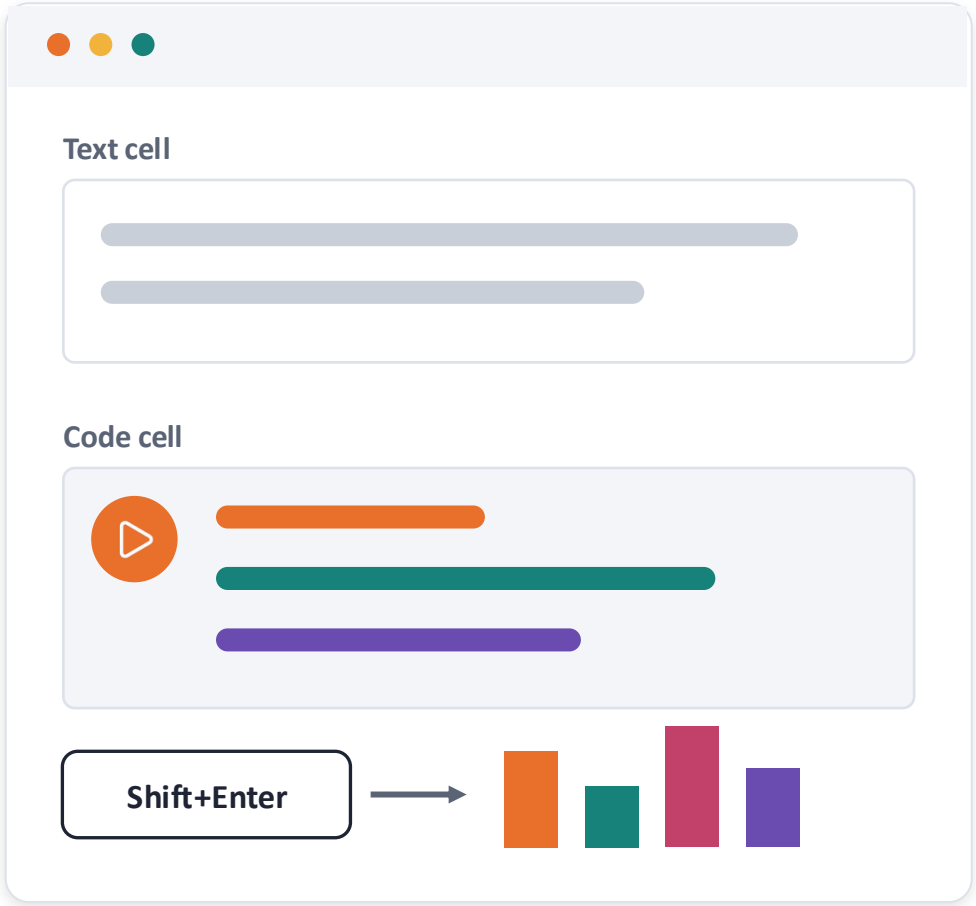
PLATFORMS · COLAB · WORKFLOW

Colab notebook basics

Open Google Colab.

- Text cells describe the question and assumptions.
- Code cells contain executable instructions.
- Shift+Enter runs a cell.
- Cells depend on earlier imports and variables.
- Saving the notebook preserves the procedure.

A run from the top helps reveal missing steps.



Files in Colab

If Drive access works, create **AI_Workshop** in My Drive and place the workshop files there.

```
from google.colab import drive
drive.mount('/content/drive')

import os
os.chdir('/content/drive/MyDrive/AI_Workshop')
print(os.listdir())
```

or

If Drive is unavailable, upload files using Colab's Files panel.

Exercise 1:

Baby Names

Annual files list each name, recorded sex (F/M), and birth count.

Combine the files into one dataset and add `year` from each filename.

First five rows of yob2025.txt:

Name	Sex	Count
Olivia	F	13,544
Charlotte	F	13,400
Emma	F	12,754
Amelia	F	12,699
Sophia	F	12,561

EXERCISE 1 · BABY NAMES

Baby names

1 Popular names:

What were the ten most popular names in your birth year, within a selected sex category?

2 Name trends:

How has your name's popularity changed over time, and how does it compare with another name?

3 Gains and losses:

Which names gained or lost the most share of recorded births between two selected years?

4 Names in both categories:

Which names are closest to a 50–50 female/male split, among names with at least 500 recorded births in a selected year?

EXERCISE 1 · BABY NAMES

Baby names

5 Are names more diverse today?

For each year and sex, compute the share of babies given one of that year's ten most popular names and the number of distinct names. Plot both over time. Are names more diverse today than in 1880?

6 Which names switched from boys to girls, or back?

For each name, compute the share of babies who were girls. Find names given mostly to boys in 1940–1959 and mostly to girls in 2005–2025, and the reverse. Include only names given to at least 1,000 babies in each period. Plot the girls' share over time for the top five.

7 Which names took off fastest?

Find the names whose share rose the most from one year to the next. Show the top ten with their year, and plot each one's share over time. What events (films, songs, public figures) might explain the jumps?

8 How have name endings changed?

Group names by their last letter and compute, for each year, the share of boys and of girls whose names end in each letter. Which endings have grown or shrunk the most since 1950? Plot the biggest changes.

Exercise 2:

AI job salaries

Historical salary records describe AI, machine-learning, and data-science jobs and their work arrangements. Each row is a reported salary record; use `salary_in_usd` for comparisons across currencies.

First five rows of ai_jobs_salaries.csv, showing selected columns:

Year	Job title	Experience code	Salary (USD)	Remote ratio
2024	Big Data Developer	MI	$117,000	0
2024	Business Intelligence Analyst	EN	$99,000	0
2024	Business Intelligence Analyst	EN	$72,000	0
2024	Data Manager	MI	$77,000	0
2024	Data Manager	MI	$49,900	0

EN = Entry · MI = Mid · SE = Senior · EX = Executive. Remote ratio: 0 = on-site, 50 = hybrid, 100 = remote.

EXERCISE 2 · AI JOB SALARIES

AI job salaries

1



Distribution:

What does the salary distribution look like, and how do the mean, median, and quartiles compare?

2



Which job titles pay the most?

For full-time jobs at US companies, group job titles that mean the same thing. Compute the median salary and number of rows for each title with at least 50 rows. Show a bar chart from highest to lowest median salary.

3



Experience:

How do median salaries and salary spreads differ across Entry, Mid, Senior, and Executive levels?

EXERCISE 2 · AI JOB SALARIES

AI job salaries

4

Are salaries rising?

For full-time jobs at US companies, exclude the partial year and compute annual median salaries overall and within each experience level. Show each year's experience mix. Why can the overall trend and the trends within levels disagree?

5

How much more do US jobs pay?

Compare median salaries at US companies with the five next most common company countries, within each experience level. Use full-time jobs. What could explain any gap besides the job itself?

6

Is remote work fading, and does it pay less?

For full-time jobs at US companies, show the share of on-site, hybrid, and remote jobs by year. Then compare each work model's median salary within each experience level.



Exercise 3,

NYC bakery transactions

Simulated purchases describe a fictional NYC bakery from October 2025 through March 2026.
Each row is one item unit; rows sharing a `Transaction` belong to the same order, or basket.

First five rows of nyc_bakery_transactions.csv:

Transaction	Item	Date/time	Period	Day type
1	Cold Brew	2025-10-01 06:48	morning	weekday
1	Oat Milk	2025-10-01 06:48	morning	weekday
1	Cold Brew	2025-10-01 06:48	morning	weekday
2	Chai Latte	2025-10-01 07:00	morning	weekday
2	Bagel with Cream Cheese	2025-10-01 07:00	morning	weekday



Task: Identify products bought together and propose bundles worth testing.

EXERCISE 3 · BAKERY TRANSACTIONS

Support, confidence, and lift

Count each product once per order, even when several units were purchased.

Suppose 20 baskets contain A in 8 cases, B in 10, and both in 6.



Measure	Calculation	Result
Support of A and B	Baskets with both ÷ all baskets	6/20 = 30%
Confidence, A to B	Baskets with both ÷ baskets with A	6/8 = 75%
Confidence, B to A	Baskets with both ÷ baskets with B	6/10 = 60%
Lift, A to B	Confidence ÷ B's basket share	0.75/0.50 = 1.5

Lift compares the conditional purchase rate with the baseline purchase rate.

Use these measures after inspecting the data and building the baskets. Market Basket Analysis Explained

EXERCISE 3 · BAKERY TRANSACTIONS

NYC bakery, read, inspect, and prepare

1. **Inspect the data.** Display the first few rows of each file. Show the number of rows, orders (`Transaction`) and distinct items, the date range, and any missing values.

2. **Clean the data.** Review the data for market basket analysis. Identify inconsistent case, trailing spaces, unnecessary punctuation, and products listed under two names. Consider dropping non-food/drink lines and modifiers that are not bought on their own. Show the issues before fixing them.

3. **Build baskets.** Create one row per order and one column per item, marking whether the item is present. Count each item once per order, even if bought more than once.

4. **Find pairs.** Compute support, confidence in both directions, and lift for each pair. Keep pairs bought together in at least 50 orders and sort by lift.

5. **Check one pair by hand.** Recompute its support, confidence, and lift step by step to confirm the table is right.

EXERCISE 3 · BAKERY TRANSACTIONS

NYC bakery, patterns and comparisons

6

Pick out the strongest pairs.

List high-lift pairs that also appear in many orders, and pairs with lift below 1 that customers buy together less often than expected. What might each suggest about the customer's visit?

7

Visualize.

Show the top pairs in a bar chart of lift and a heatmap of lift among the most popular items.

8

Look at groups of three.

Find sets of three items often bought together and describe what kind of order each looks like.

9

Compare by time.

Repeat the pair analysis for morning, afternoon, and evening, and for weekdays and weekends. Which pairs hold all day, and which only at certain times?

10

Handle seasonal items.

For items on the menu for only part of the period, analyze only dates when they were available, using `Available` in `menu_prices.csv`.

EXERCISE 3 · BAKERY TRANSACTIONS

NYC bakery, business decisions

11 **Compare by category.**
Join items to the menu's `Category` and compute lift between categories. How do the results differ from item-level pairs?

12 **Measure add-on rates.**
For each bundle idea, find the share of orders with the main item that also include the complementary product, and the number with the main item without it.

13 **Check stability.**
Compute monthly lift for the top pairs and flag those that change substantially.

14 **Estimate revenue.**
Using menu prices, propose a bundle price and estimate each bundle's monthly revenue effect under stated take-up and discount assumptions.



15 **Recommend.**
Recommend a morning combo, a lunch combo, a weekend promotion, one item to drop or reposition, and one trap: a pair that looks strong but would make a bad promotion.



CHECKS

Checks to use in every analysis

Check	Recompute a small result by hand	Reconcile totals and group counts	Inspect selected source records	Restart and run from the beginning
Workshop example	A name's share, a salary median, or a pair's lift	Birth totals, selected salary records, or eligible orders	Confirm a filter, title mapping, or cleaned basket	Confirm that saved code includes every required step

Preserve the inputs, method, and checks so the result can be reproduced.

Asking the same model to approve its own answer is not an independent check.

In Conclusion

1 Prompt Once, Compute Many Times



2 Find the code to run

